

TrES-3b

R-0515

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_240802 · created 2026-07-20T21:30:24+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460524.836 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.056
Transit depth (Rp/Rs)^2	3.84 +/- 2.2 [%]
Orbital Inclination (inc)	86.88 +/- 2.91
Airmass coefficient 1 (a1)	0.71 +/- 0.33
Airmass coefficient 2 (a2)	0.19 +/- 0.23
Scatter in the residuals of the lightcurve fit is	47.13 %
Best Comparison Star	None
Optimal Aperture	1.34
Optimal Annulus	7.25
Transit Duration (day)	0.08 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.056 vs 0.16309 (deviation 0.58σ)
Duration fit vs published	0.08 d vs 0.0567 d (deviation 2.08σ)
Mid-time O–C	3.2 min
Depth SNR	3.4
Residual scatter	47.13 %

Light curve

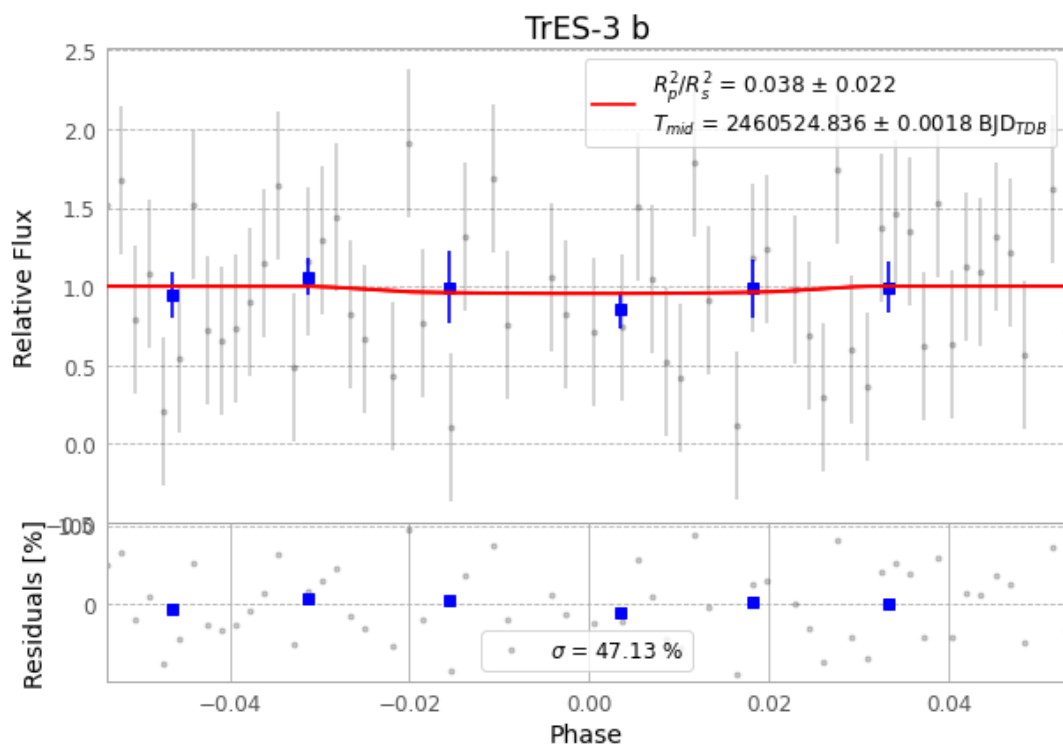


Figure 1 — FinalLightCurve_TrES-3 b_2024-08-01.png (SHA-256 40a517519471394e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [551, 202], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 37ad08ec2f149f2b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

68 FITS frames (45 MB), TRES-3240802061815.FITS → TRES-3240802093916.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-240802.log (f65c7c858420...), FinalLightCurve_TrES-3 b_2024-08-01.png (40a517519471...), FinalLightCurve_TrES-3 b_2024-08-01.csv (4344a73c3820...)

Compute resources

Wall time 190.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 21:30Z.